FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026 — ANSWER KEY (Version 1)

Biology — Class IX | Paper: SSC-I

Syllabus: Chapter 8 — Plant Physiology

Total Marks: 65 **Time: 2h 30m** **Version: 1**

SECTION A — MCQ Answer Key (12 × 1 = 12 Marks)

#	Question	Answer 1 mk	Explanation
1	Chlorosis (yellowing of leaves) occurs due to deficiency of:	**C — Magnesium**	**Magnesium** is a macronutrient required to make **chlorophyll**. Its deficiency causes **chlorosis** (yellowing). Sulphur and phosphorus deficiency cause other growth problems; calcium deficiency causes structural issues.
2	Most uptake of water and minerals from soil takes place through:	**B — Root hair**	**Root hairs** are extensions of root epidermal cells that increase surface area by approximately **67%**. They are the primary absorption sites. Root caps protect the root tip; epidermal cells do absorb but root hairs maximise the area.
3	Approximately what percentage of absorbed water is lost through stomatal transpiration?	**C — 90%**	**90%** of water loss is **stomatal transpiration** through stomata in leaves. Cuticular transpiration accounts for 7–9%; lenticular transpiration accounts for less than 3%.
4	The upward movement of water and dissolved minerals through xylem is called:	**C — Ascent of sap**	**Ascent of sap** is the upward movement of water and dissolved minerals through **xylem tissue**. Translocation is movement of organic solutes through phloem; transpiration is water loss from aerial parts.
5	According to the TACT theory, what are the four forces that move water up a plant?	**B — Transpiration pull, Adhesion, Cohesion, Tension**	The **TACT acronym** stands for **Transpiration pull, Adhesion, Cohesion, and Tension**. These four forces work together to drive the ascent of sap against gravity through xylem vessels.
6	Sugar moves through phloem mostly in the form of:	**B — Sucrose**	**Phloem sap** contains about 10–25% dry matter, most of which is **sucrose**. Glucose is the direct product of photosynthesis but is converted to sucrose before loading into phloem for transport.
7	What is the role of companion cells in translocation?	**C — Actively transport sugars into sieve tube elements**	**Companion cells** have nuclei and control the function of adjacent sieve tube elements (which lack nuclei). They **actively load sucrose** into sieve tube elements using ATP-driven carrier proteins.
8	When the rate of photosynthesis equals the rate of respiration, which gas exchange pattern occurs?	**D — Neither CO_2 nor O_2 exchanged**	This is the **compensation point of photosynthesis**, occurring at dawn and dusk. The CO_2 produced by respiration is entirely used by photosynthesis and vice versa, so **no net gas exchange** occurs with the environment.
9	Which plant group stores a small amount of water and has a thin cuticle?	**C — Mesophytes**	**Mesophytes** thrive where water is moderately available. They have a **thinner cuticle** than xerophytes and store less water than hydrophytes. Xerophytes have the thickest cuticle; hydrophytes have almost none.
10	Leaves loaded with large amounts of pigmented compounds causing them to yellow and fall are called:	**B — Excretophores**	**Excretophores** are leaves that accumulate pigmented waste compounds, turn yellow, and fall in autumn. This is a form of **excretion through leaves**. Gardeners value decomposed autumn leaves as a mineral-rich soil amendment.

11	Active transport of minerals into root cells requires energy in the form of:	**C — ATP**	**Active transport** moves minerals against the concentration gradient. It requires energy from **ATP** (adenosine triphosphate) and involves **carrier proteins** embedded in the cell membrane of root epidermal cells.
12	In the pressure flow mechanism, sugar is first loaded into sieve tubes at the:	**B — Source (leaves)**	In the **pressure flow mechanism**, companion cells actively load sucrose into sieve tube elements at the **source** (e.g., photosynthesising leaves). Water then enters by osmosis, creating hydrostatic pressure that pushes the solution toward the sink.

SECTION B — Short Answer Key (11 × 3 = 33 Marks)

2(i) Macronutrients vs Micronutrients **3 mks**

Macronutrients are needed in large amounts (greater than 0.05% dry weight). They include carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium, and magnesium. **1 mk**
Micronutrients are needed in trace amounts (less than 0.05% dry weight). They include iron, boron, manganese, copper, molybdenum, chlorine, and zinc. **1 mk**
Sixteen elements in total are essential for plant growth. Nine are macronutrients and seven are micronutrients. All are inorganic and absorbed from soil through roots. **1 mk**

OR

2(i) OR: Importance of Nitrogen **3 mks**

Nitrogen is a macronutrient absorbed from soil as inorganic nitrates (NO_3^-). It is used to make **amino acids, nucleotides, and chlorophyll**. **1 mk**
Amino acids are then used in **protein synthesis**; proteins are the most abundant organic molecules in the plant body. **1 mk**
Deficiency leads to **chlorosis** (yellowing) and severe growth retardation because chlorophyll formation is inhibited. Nitrogen fertilisers are added to nitrogen-depleted soils to correct this. **1 mk**

2(ii) Internal Structure of Root **3 mks**

From outermost to innermost, the root layers are: **epidermis (epiblema) → cortex → endodermis → pericycle → vascular bundle (xylem and phloem)** → pith (in monocots). **1 mk**
Casparian strips are waxy (suberin) deposits in the radial walls of endodermal cells. They prevent uncontrolled movement of water and solutes, forcing all materials to pass through the cell membrane, allowing the plant to regulate what enters the xylem. **1 mk**
Vascular bundles in **monocot roots** form a ring with central pith; in **dicot roots** they are star-shaped in the centre without pith. **1 mk**

OR

2(ii) OR: Role of Root Hairs **3 mks**

Root hairs are extensions of root epidermal cells that penetrate spaces between soil particles. They increase the surface area of the root by approximately **67%**. **1 mk**
They are the primary sites for **absorption of water and minerals**. Water enters by **osmosis**; minerals enter by diffusion, facilitated diffusion, or active transport. **1 mk**
After entering root hairs, absorbed substances pass through cortex, endodermis, and pericycle before reaching the **xylem cells** for upward transport. **1 mk**

2(iii) Passive vs Active Transport in Mineral Uptake 3 mks

Passive transport includes diffusion and facilitated diffusion. Minerals move **along** their concentration gradient (high to low concentration) with no energy expenditure. Carrier proteins help in facilitated diffusion. **1 mk**

Active transport moves minerals **against** the concentration gradient, from low to high concentration inside root cells. It requires **ATP energy** and specific carrier proteins; it is selective. **1 mk**

Active transport is essential when mineral concentration inside root cells is higher than in soil solution, allowing the plant to accumulate minerals it needs. Both mechanisms together ensure the plant obtains all required nutrients. **1 mk**

OR

2(iii) OR: Water Absorption Mechanism 3 mks

Water absorption by roots occurs by **osmosis**, a passive transport mechanism. Water moves from a region of high concentration (soil) to low concentration (inside root cells) across a **selectively permeable** membrane. **1 mk**

If water moves into a cell it is **endosmosis**; if it moves out it is **exosmosis**. The cell wall allows water and minerals to pass freely but the cell membrane is selectively permeable. **1 mk**

Path of water: soil → root hair cell → cortex → endodermis → pericycle → **xylem cells**. From xylem, water is pulled upward toward the leaves by transpiration. **1 mk**

2(iv) Three Types of Transpiration 3 mks

(i) **Stomatal transpiration**: water vapour lost through **stomata** (tiny pores mainly on leaves). Accounts for **90%** of all water loss. Stomata are open during the day. **1 mk**

(ii) **Cuticular transpiration**: water lost through the **cuticle** (waxy layer) of the general leaf surface. Accounts for **7–9%** of water loss. **1 mk**

(iii) **Lenticular transpiration**: water lost through **lenticels** in the stem. Lenticels are scar-like regions with ruptured epidermis and loosely packed cortical cells. Accounts for **less than 3%** of water loss. **1 mk**

OR

2(iv) OR: Transpiration as Necessary Evil 3 mks

Advantages: (1) Creates the transpiration pull that drives **ascent of sap** (water and minerals to leaves). (2) Provides **cooling** to the plant through evaporative cooling. **2 mks**

Disadvantage: Plants lose approximately 99% of absorbed water through transpiration. Excessive unchecked transpiration causes **wilting** (loss of turgor) and can ultimately cause plant death. That is why transpiration is called a “necessary evil”: without it the plant cannot survive, yet it causes massive water loss. **1 mk**

2(v) Effect of Temperature on Transpiration 3 mks

Higher temperature lowers **humidity in the air**, increasing the rate of evaporation from mesophyll cell surfaces. For every **10°C rise**, the transpiration rate roughly **doubles**. **1 mk**

At very high temperatures (**40–45°C**), stomata **close** to prevent excessive water loss. This protects the plant from wilting. **1 mk**

If high temperatures persist and soil water is insufficient, the plant begins to **wilt and may die**. Stomatal closure reduces photosynthesis as CO_2 entry is also restricted. **1 mk**

OR

2(v) OR: Effect of Wind on Transpiration 3 mks

Wind is **moving air** that speeds up the **diffusion** of water molecules away from the leaf surface. This removes the water vapour boundary layer around the leaf. **1 mk**

As water vapour is removed, the concentration gradient between the air inside the leaf (humid) and outside (drier) increases, so water diffuses outward faster, increasing evaporation from **mesophyll cell surfaces**. **1 mk**

When air is **calm and still**, diffusion slows as water vapour accumulates around the leaf, creating a saturated boundary layer that reduces the rate of transpiration. **1 mk**

2(vi) Cohesion and Tension in TACT Theory 3 mks

Cohesion is the attraction between water molecules themselves. Because water is a **polar molecule**, **hydrogen bonds** form between neighbouring water molecules, holding them together like a string. 1 mk
Tension is the pulling force created in the water column when transpiration removes water from the top. This tension travels down through the continuous water column in the xylem. 1 mk
Together, cohesion keeps molecules linked and tension transmits the pull downward so the entire column moves as a single unit. Imagine the water column as a steel wire: as long as transpiration continues, the wire remains taut and is pulled upward. 1 mk

OR

2(vi) OR: Adhesion in Water Transport 3 mks

Adhesion is the attraction between water molecules and the walls of xylem cells. Both water and **cellulose** (in cell walls) are polar molecules. 1 mk
Water molecules are attracted to and "stick to" the xylem cell walls. This adhesion helps water **move upward against gravity** by resisting the tendency of the column to fall back downward. 1 mk
Adhesion also keeps water in the xylem even when transpiration temporarily slows, preventing the column from draining back into the root. It is especially important in **tall plants** where the gravitational pull is strong. 1 mk

2(vii) Translocation: Tissue, Direction, Source and Sink 3 mks

Translocation is the transport of prepared food (organic solutes, mainly sucrose) to different parts of the plant through **phloem** tissue. 1 mk
Movement is not unidirectional. It goes from a **source** (a region that supplies food, e.g., photosynthesising leaves or storage organs) to a **sink** (a region that uses food, e.g., roots, fruits, growing tips). 1 mk
Leaves and storage organs are sources; roots, fruits, and stem apices are sinks. Stems can act as both. The direction of flow is determined by the relative pressures at source and sink. 1 mk

OR

2(vii) OR: Phloem Sap Composition and Sieve Tube / Companion Cells 3 mks

Phloem sap (translocating fluid) contains about 10–25% dry matter. Most of this is **sucrose**, with other organic compounds in smaller amounts. 1 mk
Sieve tube elements are long tube-like structures that transport organic solutes throughout the plant. They **lack nuclei**, so they cannot control their own activities. 1 mk
Companion cells sit next to sieve tube elements, have nuclei, and control the function of adjacent sieve elements. They actively load sucrose into sieve tubes using ATP. Together they form the conducting channels for organic solutes. 1 mk

2(viii) Gas Exchange in Leaves During Daytime 3 mks

During the day, plants carry out both **photosynthesis** and respiration. The rate of photosynthesis is greater than respiration in daylight. 1 mk
Photosynthesis uses more **CO_2** than respiration produces, so plants draw in extra CO_2 from the environment through open stomata. Photosynthesis produces more **O_2** than respiration needs, so excess oxygen is released. 1 mk
The overall result: leaves **absorb CO_2 and release O_2** during the day. Gas exchange occurs primarily through **stomata** on leaf surfaces and through **lenticels** on stems. 1 mk

OR

2(viii) OR: Compensation Point of Photosynthesis 3 mks

The **compensation point** is the light intensity at which the rate of photosynthesis exactly equals the rate of respiration. 1 mk
It occurs at **dawn and dusk**, when light intensity is low. At this point, all CO_2 produced by respiration is consumed by photosynthesis, and all O_2 produced by photosynthesis is consumed by respiration. 1 mk
Therefore there is **no net gas exchange** with the environment. This is a transitional state between the daytime pattern (absorb CO_2, release O_2) and the night pattern (absorb O_2, release CO_2). 1 mk

2(ix) Homeostasis in Plants **3 mks**

Homeostasis in plants refers to the adjustment and adaptation of plants to varying degrees of environmental temperatures and availability of water and salts in the soil. **1 mk**

It is essential because: (1) it allows plants to respond to **temperature changes** and survive in various climates; (2) it maintains **adequate hydration levels**; (3) it enables response to stressors such as drought, salinity, pathogens, and herbivores. **1 mk**

Homeostasis helps plants activate **defence mechanisms**, adapt to changing environmental conditions, and maintain internal stability, which contributes to their ecological success. **1 mk**

OR

2(ix) OR: Plant Excretion and Excretophores **3 mks**

Plants excrete (1) **carbon dioxide** (waste of respiration, released at night) and **oxygen** (waste of photosynthesis, released in daytime) through gas exchange. (2) **Water** is excreted through transpiration as a waste of both photosynthesis and respiration. **1 mk**

Excretophores are leaves loaded with large amounts of pigmented waste compounds. They turn yellow and fall from the plant, carrying waste materials away from the plant body. This is excretion through leaves. **1 mk**

Decomposed autumn leaves are mineral-rich because the waste compounds they contain break down and release minerals back into the soil, which is why **gardeners use autumn leaves** as a natural fertiliser. **1 mk**

2(x) Hypotonic vs Hypertonic Situations **3 mks**

Hypotonic situation: The external environment contains more water and less dissolved solutes than inside the cell. Water moves into the cell by osmosis (**endosmosis**), making the cell swell and become firm (**turgid**). **1 mk**

Hypertonic situation: The external environment has less water and more dissolved solutes than inside the cell (more concentrated than cell sap). Water moves out (**exosmosis**), causing the cell to shrink and become limp (**flaccid**). **1 mk**

Both situations require osmotic adjustments to maintain balance. In hypotonic conditions the plant gains turgor pressure, useful for structural support; in hypertonic conditions (e.g., drought or high salinity) the plant can wilt if no adjustment occurs. **1 mk**

OR

2(x) OR: Osmotic Adjustment and Isotonic Situation **3 mks**

Osmotic adjustment (osmoregulation) is the plant's mechanism for maintaining the correct balance of water and solutes in its body. It creates three possible situations: hypotonic, hypertonic, and isotonic. **1 mk**

In an **isotonic situation**, the water and solute concentrations inside and outside the cell are equal. There is no net movement of water in or out of the cell, so the cell is in balance. **1 mk**

The isotonic situation is considered the ideal state for a cell but is **rarely found in nature** because environmental conditions constantly fluctuate, constantly shifting the balance between hypotonic and hypertonic. **1 mk**

2(xi) Hydrophytes vs Xerophytes **3 mks**

Hydrophytes live in water-rich environments. They have the **highest transpiration rate**; stomata are on the **upper leaf surface**; stomata remain open day and night; cuticle is almost absent. **1 mk**
Xerophytes are adapted to extremely dry conditions. They have the **lowest transpiration rate**; stomata are **sunken** (recessed into the leaf surface to reduce air movement); stomata open only at **night**; cuticle is very thick. **1 mk**
Xerophytes (succulents) also have high water storage capacity in their tissues. These contrasting adaptations reflect the opposite water availability challenges each group faces. **1 mk**

OR

2(xi) OR: Gas Exchange in Roots **3 mks**

Roots are non-photosynthetic parts of the plant. They exchange gases with soil air through **root epidermal cells**. They absorb **oxygen** for cellular respiration and release **carbon dioxide** as a waste product. **1 mk**
This pattern **does not change** between day and night because roots never carry out photosynthesis. Leaves change their gas exchange pattern depending on light intensity, but roots always absorb O_2 and release CO_2. **1 mk**
Gas exchange in roots occurs through the root cell surfaces and through intercellular air spaces. In waterlogged soils, restricted air access can cause oxygen deficiency (anaerobic conditions), harming root respiration. **1 mk**

SECTION C — Long Answer Key (4 × 5 = 20 Marks)

3(i) Mineral Nutrition in Plants **5 mks**

Nutrient: Any substance that provides necessary elements to the organism for growth and metabolism. Plant nutrients are mostly inorganic, absorbed from the environment. **1 mk**

Macronutrients vs Micronutrients: 16 elements are essential. Nine are macronutrients (needed in amounts > 0.05% dry weight): carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium, and magnesium. Seven are micronutrients (needed in trace amounts < 0.05% dry weight): iron, boron, manganese, copper, molybdenum, chlorine, and zinc. **1 mk**

Role of Nitrogen: Absorbed as nitrates (NO_3^-) from soil. Used to make amino acids, nucleotides, and chlorophyll. Amino acids build proteins, the most abundant organic constituent of the plant. Nitrogen deficiency causes **chlorosis** (yellowing) and severe growth retardation. Added in nitrogen fertilisers for depleted soils. **1.5 mks**

Role of Magnesium: Absorbed as Mg^{2+} ions. Used to make **chlorophyll** (magnesium is the central atom in the chlorophyll molecule). Magnesium deficiency also causes **chlorosis**. Added in dolomitic limestone or magnesium sulphate fertilisers for depleted soils. **1.5 mks**

OR

3(i) OR: Internal Structure of Root and Root Hairs — Mineral and Water Uptake **5 mks**

Root Layers (outer to inner): epidermis (epiblema) → cortex → endodermis → pericycle → vascular bundle (xylem and phloem). Monocot roots have pith at the centre; dicot roots do not. **1 mk**

Root Hairs: Extensions of epidermal cells that penetrate spaces between soil particles. They increase surface area by ~67% and are the primary absorption sites for water and minerals. **1 mk**

Casparian Strip: A waxy (suberin) deposit in the radial walls of endodermal cells. Forces all materials entering the xylem to pass through the endodermal cell membrane, allowing selective regulation. **1 mk**

Passive Mineral Uptake: Diffusion along concentration gradient; facilitated diffusion using membrane carrier proteins. No energy required. Minerals move from high concentration in soil to lower concentration in root cells. **1 mk**

Active Mineral Uptake: Against concentration gradient, requires **ATP**. Selective; allows plant to accumulate essential minerals even when soil concentration is low. Water absorption is by **osmosis** (passive), moving from high water concentration in soil to lower concentration in root cells. Path: soil → root hair → cortex → endodermis → pericycle → xylem. **1 mk**

3(ii) TACT Theory — Full Explanation with Water Path **5 mks**

The **TACT theory** explains the ascent of sap (upward movement of water and dissolved minerals) through xylem. Four forces work together:

T — Transpiration Pull: When stomata open, water evaporates from leaf mesophyll cells into air spaces and out through stomata. This creates a **suction force** (pulling force) that draws water upward through xylem from roots to leaves. It is driven by solar energy. **1 mk**

A — Adhesion: Attraction between water molecules and the cellulose walls of xylem cells (both polar). Adhesion helps water climb the walls and prevents the water column from falling back under gravity. **1 mk**

C — Cohesion: Attraction between water molecules themselves through hydrogen bonds. Keeps water molecules linked as a continuous unbroken chain. Imagine a string of beads: pulling the top bead (transpiration) pulls all others up because cohesion holds them together. **1 mk**

T — Tension: The tension (mechanical pulling force) created in the water column by transpiration. As long as transpiration continues, the water column remains taut and is pulled upward as a single unit. **1 mk**

Complete water path:

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Soil water ↓ (osmosis) Root hair cell ↓ (osmosis through cortex, endodermis, pericycle) Xylem vessel in root ↓ (TACT forces) Xylem vessel in stem ↓ (TACT forces) Xylem vessel in leaf vein ↓ (diffusion) Mesophyll cell ↓ (evaporation through open stomata) Atmosphere
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1 mk

OR

3(ii) OR: Transpiration — Types, Stomatal Regulation, and Necessary Evil **5 mks**

Definition: Transpiration is the loss of water in the form of vapour from the aerial parts of the plant. Approximately 99% of water absorbed by roots is lost this way. **0.5 mk**

Three Types:
(1) **Stomatal transpiration**: through stomata; 90% of loss. (2) **Cuticular transpiration**: through cuticle; 7–9% of loss. (3) **Lenticular transpiration**: through lenticels in stem; less than 3% of loss. **1.5 mks**

Stomatal Opening and Closing: Stomata are mainly present on leaf surfaces. Plants with large leaf surface area (more stomata) transpire more. Stomata are widely open during the day and nearly close at night, so transpiration is higher during daytime. At 40–45°C stomata close to conserve water. **1.5 mks**

Necessary Evil Evaluation:
Advantages: (1) Creates transpiration pull, the driving force for the ascent of sap, delivering water and minerals to all leaf cells. (2) Provides evaporative **cooling**. (3) Facilitates CO_2 entry through open stomata for photosynthesis.
Disadvantage: Massive water loss (99% of absorbed water). Excessive transpiration causes wilting and plant death. Thus transpiration is “necessary” (plant cannot function without it) yet “evil” (causes damaging water loss). **1.5 mks**

3(iii) Pressure Flow (Mass Flow) Mechanism of Translocation **5 mks**

Step 1 — Loading at Source: Sugars (sucrose) produced in leaves (source) are **actively loaded** into sieve tube elements by companion cells using ATP-driven carrier proteins. This raises sucrose concentration in the phloem. **1 mk**

Step 2 — Water Entry at Source: The high solute concentration in the phloem reduces water potential. Water moves in from nearby xylem cells by **osmosis**, increasing **hydrostatic (turgor) pressure** in the phloem at the source end. **1 mk**

Step 3 — Flow from Source to Sink: The high pressure at the source pushes the sugary solution through sieve tubes toward the lower-pressure **sink** (roots, fruits, storage organs), like water pushed through a hose. This bulk movement is translocation. **1 mk**

Step 4 — Unloading at Sink: At the sink, companion cells **actively unload sucrose** out of the sieve tube (requiring ATP). The loss of solutes increases water potential in the phloem at the sink. Water leaves the phloem by osmosis back into xylem or surrounding cells. **1 mk**

Step 5 — Result: This loss of water at the sink reduces hydrostatic pressure there, maintaining the pressure gradient between source and sink that drives continued flow. The system is thus like a **pressurised loop**. Sieve tube elements lack nuclei; companion cells next to them provide metabolic control. **1 mk**

OR

3(iii) OR: Xylem vs Phloem Transport — Comparison Table **5 mks**

Feature	Xylem Transport (Ascent of Sap)	Phloem Transport (Translocation)
Materials transported	Water and dissolved minerals (inorganic)	Organic solutes: mainly sucrose; also amino acids
Direction of flow	Unidirectional: upward only (root to leaf)	Bidirectional: source to sink (any direction)
Cell types involved	Tracheids and vessel elements (dead, no nuclei)	Sieve tube elements (no nuclei) + companion cells (with nuclei)
Energy requirement	No direct ATP for flow; passive (TACT forces)	Active loading and unloading requires ATP
Driving force	Transpiration pull, adhesion, cohesion, tension	Hydrostatic pressure gradient (pressure flow)
Where driven from	Driven from the top (transpiration at leaves)	Driven from the source (high pressure) to sink (low pressure)

Award 1 mark per correct row (any 5 rows fully correct = full marks). **5 mks**

3(iv) Gas Exchange in Plants — Day, Night, Compensation Point, Leaves and Roots **5 mks**

Gas exchange defined: The process by which a plant exchanges gases (O_2 and CO_2) with its environment. It occurs primarily through **stomata** in leaves and through **lenticels** in stems. **0.5 mk**

Pattern in leaves during the day: Both photosynthesis and respiration occur. Rate of photosynthesis > rate of respiration. Net result: leaves **absorb CO_2** from air (for photosynthesis) and **release O_2** (excess from photosynthesis). **1 mk**

Pattern in leaves at night: No photosynthesis. Only respiration. Leaves **absorb O_2** and **release CO_2**, exactly like animals. **1 mk**

Compensation point (dawn and dusk): Light intensity is low; rate of photosynthesis equals rate of respiration. CO_2 produced by respiration is entirely consumed by photosynthesis; O_2 produced by photosynthesis is entirely consumed by respiration. **No net gas exchange** with the environment. **1 mk**

Pattern in roots (non-photosynthetic): Roots always absorb **O_2** and release **CO_2**, day and night, because there is no photosynthesis in roots. Gas exchange here occurs through root epidermal cells and intercellular air spaces. **1 mk**

Summary table:

Part	Condition	Gas Absorbed	Gas Released
Leaves	Daytime	CO_2	O_2
Leaves	Nighttime	O_2	CO_2
Leaves	Dawn/Dusk (comp. point)	Neither	Neither
Roots	Day and night	O_2	CO_2

0.5 mk

OR

3(iv) OR: Osmotic Adjustment — Three Situations and Hydrophyte/Mesophyte/Xerophyte Adaptations **5 mks**

Osmotic adjustment (osmoregulation): The plant's mechanism for maintaining the right balance of water and solutes in its body, creating three situations. **0.5 mk**

Hypotonic situation: External environment has more water and fewer solutes than inside the cell. Water enters by osmosis (endosmosis); cell becomes **turgid** (firm and swollen). **1 mk**
Hypertonic situation: External environment has less water and more solutes than inside the cell. Water exits by osmosis (exosmosis); cell becomes **flaccid** (limp and shrunken). **1 mk**
Isotonic situation: Water and solute concentrations inside and outside the cell are equal. No net water movement; the cell is in balance. Rarely found in nature due to constant environmental fluctuations. **0.5 mk**

Adaptations comparison:

Feature	Hydrophytes (water-rich)	Mesophytes (moderate water)	Xerophytes (water-poor)
Transpiration rate	Highest	Moderate	Lowest
Stomatal placement	Upper leaf surface	Lower leaf surface	Sunken stomata
Stomatal behaviour	Open day and night	Open during day	Open only at night
Cuticle thickness	Almost none	Thin	Very thick
Water storage	Minimal	Moderate	High (succulents)

2 mks